

(a) Today — HiCache

(b) Ours — KV-aware unified memory

(c) Placement policy

ranked by bandwidth measured at boot — not by tier

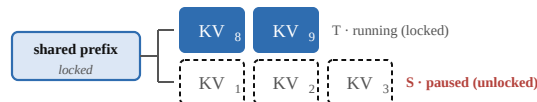
Control plane

no runtime placement

host tier sized once at boot (~hicache-ratio)
61 GB held whether the cache is 5% or 95% full;
write policy and storage backend do not change it

Prefill instance (GPU) — radix cache

KV pool 83% full — but only 1% live; 81% is evictable cache



no path to peer HBM ❌

Decode instance (GPU)

14% used, all of it live — holds no cache, 86% free



capacity already paid for, and left unused

evict → host, always ↓

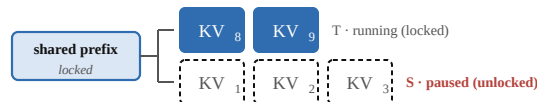
Host DRAM — HiCache

61 GB reserved at boot



Prefill instance (GPU) — radix cache

KV pool 83% full — but only 1% live; 81% is evictable cache



fetch / re-park — peer copy, 52.8 GB/s = 4.0× host 3 ↑ 4 ↓

Decode instance (GPU)

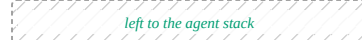
14% used, all of it live — holds no cache, 86% free



priority 3 — only when no GPU slab fits ↓

Host DRAM

empty — nothing parked here



1 idlest NVLink peer GPU HBM

52.8 GB/s

headroom ≥ size; 4.0× the host round trip

2 any GPU whose link beats host

topology-ranked, re-measured per node

non-NVLink peer GPU

3.3 GB/s

excluded — slower than host DRAM

3 host DRAM overflow

26.3 GB/s × 2

last resort — and the host path pays PCIe twice

4 drop → recompute

1203 ms @ 8k

victim cache: correctness never depends on it

serving always wins

on pressure, the lowest-reuse parked KV moves GPU → GPU, then → host, then is dropped (§3.3). Parked KV is a victim cache: the consistency fallback is recompute, so borrowed memory can always be handed back.

Why not CUDA Unified Memory? (§2.2)

- eviction has exactly one destination — host. UM grows the very commitment this design sets out to reduce.
- it cannot cross process boundaries, and 2P+2D is four separate processes.
- it cannot discard: managed pages must be preserved, but KV may be dropped because recompute is always correct.

- 1 The next turn of session S arrives; the prefill node's local radix misses — its prefix was evicted under pressure while the session was in a tool call.
- 2 The control plane resolves the prefix hash: the park index says loc = GPU2, and the link table ranks that copy above host DRAM.
- 3 Bulk peer-GPU copy GPU2 → GPU0, inserted into the radix as an ordinary prefix hit. No page faults, no host round trip.
- 4 Prefill computes only the new tokens and re-parks the extended prefix on whichever GPU is idlest then — host DRAM is never touched.

■ serving (current batch)
 ■ waiting — session paused between turns
 ■ completed
 evicted / absent
 capacity held but unused

The cache cannot tell waiting from completed — both are LRU-evictable entries ordered by last access. That is what parking is for.